AIDOOi

AID-LOG-002

LIVING ROOM COMPOSITION, FIT & BUDGET LOGIC

Revision 1 | MVP

Purpose: Define how AIDOOi converts Living Room dimensions into a complete initial furniture composition, preferred product fit, room-level budget range, and dynamically recalculated final recommendation price.

1. Scope and Standard Role

AID-LOG-002 is the backend decision standard for the Living Room customer path. It consumes the room dimensions captured in AID-SPC-002 and uses AID-LOG-001 product price intelligence to build an appropriate complete initial Living Room recommendation.

This standard defines four connected layers: **Room Size -> Composition -> Product Fit -> Budget & Recommendation**.

It does not redefine product taxonomy, product price ranges, style, colour, inventory, or the customer question flow. Those remain controlled by the relevant CAT, DATA, SPC and LOG standards.

2. Governing Principles

- The customer provides Living Room Length and Width before budget selection.
- AIDOOi calculates room area and assigns a system Room Size Classification. The customer does not manually choose the size class.
- AIDOOi initially generates a complete room composition appropriate to that size class.
- The customer may remove individual products after recommendation. The final composition price is recalculated dynamically.
- Default Composition is not a hard constraint. Layout and actual available space may justify a different valid composition.
- Preferred product dimensions are guidance for matching, not absolute cut-off filters.
- Room area alone is not sufficient for final fit. Room dimensions/shape and available placement space must also be considered.
- Room Budget Tier is a target for the complete composition; it does not require every product to belong to the same product-level tier.

3. Living Room Size Classification

Calculated Area	Classification
< 12 m²	Compact
12-20 m²	Small
20-30 m²	Medium
30-45 m²	Large
45+ m²	Extra Large

Calculation: **Length x Width = Room Area**. The size class activates the default composition and preferred-fit profile below.

4. Size-Based Default Composition and Fit

4.1 Compact Living Room - < 12 m²

Product	Default Qty	Preferred Fit Logic
Sofa	1	2-Seater; approx. 140-180 cm width
Accent Chair / Armchair	0	Not included by default
Coffee Table	1	Compact; approx. 60-90 cm
Side / End Table	1	Approx. 30-45 cm
TV Unit / Media Console	1	Approx. 100-140 cm
Rug	1	Small; approx. 120x180 cm
Floor / Table Lamp	1	Compact footprint; layout dependent
Decor	2	Selected from available placement zones

4.2 Small Living Room - 12-20 m²

Product	Default Qty	Preferred Fit Logic
Sofa	1	2-3 Seater; approx. 170-220 cm width
Accent Chair / Armchair	1	Compact/Medium; approx. 60-80 cm width
Coffee Table	1	Approx. 70-110 cm

Product	Default Qty	Preferred Fit Logic
Side / End Table	1	Approx. 35-50 cm
TV Unit / Media Console	1	Approx. 120-180 cm
Rug	1	Medium; approx. 160x230 cm
Floor / Table Lamp	1	Layout dependent
Decor	3	Placement-zone driven mix

4.3 Medium Living Room - 20-30 m²

Product	Default Qty	Preferred Fit Logic
Sofa	1	Default 3-Seater; approx. 200-250 cm
Accent Chair / Armchair	1	Approx. 70-90 cm width
Coffee Table	1	Approx. 90-130 cm
Side / End Table	1-2	Approx. 40-55 cm
TV Unit / Media Console	1	Approx. 160-220 cm
Rug	1	Large; approx. 200x300 cm
Floor / Table Lamp	1-2	Layout dependent
Decor	3-4	Placement-zone driven mix

4.4 Large Living Room - 30-45 m²

Product	Default Qty	Preferred Fit Logic
Sofa	1	4-Seater / Sectional; approx. 240-300 cm
Accent Chair / Armchair	2	Approx. 70-95 cm width each
Coffee Table	1	Approx. 110-150 cm
Side / End Table	2	Approx. 40-60 cm
TV Unit / Media Console	1	Approx. 180-260 cm
Rug	1	Large / XL; approx. 200x300 to 250x350 cm
Floor / Table Lamp	2	According to seating/layout
Decor	4-6	Placement-zone driven mix

4.5 Extra Large Living Room - 45+ m²

Product	Default Qty	Preferred Fit Logic
Sofa	1	5+ Seater / Sectional; approx. 280-360+ cm
Accent Chair / Armchair	2-3	Approx. 75-100 cm width each
Coffee Table	1-2	Approx. 120-160 cm or complementary pair
Side / End Table	2-3	Approx. 40-65 cm
TV Unit / Media Console	1	Approx. 220-300+ cm
Rug	1	Extra Large; approx. 250x350 cm+
Floor / Table Lamp	2-3	According to seating zones
Decor	5-8	Placement-zone driven mix

5. Decor Placement Logic

Decor quantity is controlled by the size profile, but product type should be selected according to actual placement opportunities rather than a fixed shopping list.

Available Placement Zone	Eligible Decor Direction
Wall zone	Wall Art / Mirror

Available Placement Zone	Eligible Decor Direction
Console or table surface	Vase / Decorative Object / Sculpture / Candle Holder
Open floor or corner	Planter / Floor Vase or other suitable floor decor

If a room has limited usable surfaces, AIDOOi should shift decor toward wall or floor zones rather than forcing surface accessories. Decor dimensions must remain compatible with the selected placement zone.

6. Extra Large Zoning Rule

For 45+ m² Living Rooms, AIDOOi may create a secondary furniture group or zone when the room shape, dimensions and available layout support it. Examples include a Main Seating Area plus a Secondary Conversation or Reading Area.

A second zone is **not automatic**. A large area alone is insufficient justification, and the system should not add unnecessary products or cost solely because the room exceeds 45 m².

7. Living Room Budget Tiers

Room Size	Essential	Value	Standard	Premium	Luxury
Compact <12	800-2,000	2,000-3,500	3,500-6,000	6,000-10,000	10,000+
Small 12-20	1,500-3,500	3,500-6,000	6,000-10,000	10,000-17,000	17,000+
Medium 20-30	2,500-5,000	5,000-8,500	8,500-14,000	14,000-23,000	23,000+
Large 30-45	4,000-7,500	7,500-12,000	12,000-20,000	20,000-33,000	33,000+
Extra Large 45+	6,000-10,000	10,000-16,000	16,000-27,000	27,000-45,000	45,000+

All ranges are AED and represent the target budget for the **initial complete composition** generated for that Living Room size.

8. Budget Allocation Logic

- The customer selects one room-level tier: Essential, Value, Standard, Premium or Luxury.
- The selected room tier activates a total target range for the applicable Living Room size.
- AIDOOi uses AID-LOG-001 to source product-level price options for each required composition slot.
- The engine may mix product-level tiers to create the strongest overall composition within the selected room range.
- Higher-cost anchor products such as Sofa and Accent Chair may absorb a larger share of the room budget; accessories do not need to mirror the room tier.
- Fit, Style, Colour, availability and the total room budget must all remain satisfied when ranking candidate compositions.

9. Selected Room Budget vs Final Composition Price

Selected Room Budget Tier is used to generate the initial complete recommendation. **Final Composition Price** is the sum of the Current Selling Prices of the products that remain in the customer's composition.

After recommendation, the customer may remove products such as a Rug, TV Unit, Accent Chair or Decor item. The system must immediately recalculate the total. The revised final price is not required to remain inside the original room-tier range.

Formula: Final Composition Price = Sum of Current Selling Price of all selected/remaining product offers.

10. Recommendation Sequence

Step	System Action
1	Receive Living Room Length and Width from AID-SPC-002.
2	Calculate area and assign Compact / Small / Medium / Large / Extra Large.
3	Load the default composition and preferred-fit profile for that size.
4	Refine fit using room dimensions/shape and available placement space.
5	Load the five applicable Living Room budget ranges.
6	Customer selects a room-level budget tier.
7	Use AID-LOG-001 plus Style, Colour and availability standards to assemble candidate compositions.
8	Rank and present a complete Living Room recommendation.

Step	System Action
9	Allow product removal and dynamically recalculate Final Composition Price.

11. Relationship to Other Standards

Reference	Role
AID-SPC-002	Living Room Path - captures dimensions, optional photos/floor plan, style, colour and budget journey.
AID-LOG-001	Product Price Range Standard - central source for individual product price intelligence.
AID-DATA-007	Product Colour Standard.
AID-DATA-009	Product Style Standard.
AID-DATA-010	Product Pricing Standard - Current Selling Price and offer-level pricing.
AID-DATA-011	Product Availability & Inventory Standard.
AID-CAT-001 to AID-CAT-008	Relevant product taxonomy and practical attributes.

12. Governance and MVP Boundaries

- The size thresholds, quantities, preferred dimensions and room budget ranges in this revision are approved MVP configuration values.
- Preferred dimensions are not hard filters and may be overridden by validated layout fit.
- Default Composition is not a mandatory shopping list; it is the initial complete recommendation profile for that size.
- The customer is not asked to manually choose the system Room Size Classification.
- Future versions may refine composition, fit and budget ranges using live inventory, room-layout performance, conversion data and customer behavior.
- Bedroom, Dining Room, Home Office, Studio Apartment, Full Apartment and Villa logic must be maintained as separate LOG standards.

13. Standard Summary

AID-LOG-002 converts a Living Room's measured dimensions into a size-aware, complete initial furnishing composition. It determines the default quantity and preferred fit of products, activates a size-specific total budget range, uses AID-LOG-001 to select real purchasable products, and recalculates the final room price when the customer changes the composition. This keeps the customer journey simple while preserving structured backend intelligence.